Rabee Al-Qasem at AraGenre 2026: Conformer-Based Hierarchical Classifier for Arabic Genres

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Abstract

This paper presents the first investigation into the performance of the Conformer architecture for hierarchical text classification, specifically applied to predicting two-level Arabic text genres. The proposed system combines multilingual sentence embeddings, target label definition representations, a Conformer projection module, an auxiliary broad-genre classifier, synthetic data augmentation, and seed-based ensembling. Evaluated on the official shared task, the approach achieved a Hierarchical Macro-F1 of 0.4399, demonstrating strong high-level genre classification (Broad Macro-F1 of 0.6288) while highlighting ongoing challenges in distinguishing semantically close specific subcategories (Specific Macro-F1 of 0.2511).

1 Introduction

Hierarchical classification (**HC**) is an advanced form of classification in which classes or labels are organized in a tree-like structure, consisting of main classes and subclasses that represent parent-child relationships (Silla Jr and Freitas, 2011). This setting is challenging because errors at one level can affect predictions at subsequent levels (Banerjee et al., 2019). In top-down approaches, a misprediction at a higher level may prevent the model from reaching the correct lower-level category, leading to errors across the hierarchy (Zhou et al., 2020; Wang et al., 2011; Mao et al., 2019).

Hierarchical datasets may also suffer from class imbalance, especially at the child level, where some fine-grained categories contain significantly fewer training examples than others. The hierarchy itself may be unbalanced, with some parent categories containing many subcategories while others contain only a few. These issues make it more difficult for the model to learn reliable representations for rare and fine-grained classes (Zhu et al., 2023; Huang et al., 2022). Another challenge is the semantic similarity between sibling labels, which may result in confusion between closely related subcategories (Song et al., 2023; Xiong et al., 2024). Furthermore, the model must maintain hierarchical consistency by ensuring that the predicted child category is associated with the correct parent category (Liu et al., 2023). Evaluation is also challenging because errors within the same branch may be less severe than errors involving categories from different branches (Plaud et al., 2024).

To the best of our knowledge, this work is the first to investigate the Conformer architecture for hierarchical text classification in Arabic. By evaluating its ability to capture both local and global dependencies in textual representations, we provide an initial assessment of its suitability for this task.

2 Shared Task Background

The proposed objective of the shared task is to develop an AI solution to predict the communicative genre of Arabic texts across two levels (Ferhat et al., 2024): Broad Genre, which refers to high-level communicative functions such as Informative, Interactive, Creative, Religious, Legal, or Learning, and Specific Genre, which refers to finer-grained genre subtypes such as encyclopaedic writing, forum discussion, educational explanation, diplomatic communication, or religious commentary. Therefore, the organizers of the shared task formulated this problem as an HC problem, in which each specific genre is associated with a general genre to construct the parent–child relationship (El-Haj et al., 2026). Model performance is evaluated using several related metrics, including Hierarchical Macro-F1, Specific Macro-F1, Specific Weighted-F1, Specific Accuracy, Broad Macro-F1, Broad Weighted-F1, and Broad Accuracy (Kosmopoulos et al., 2015). The provided dataset consists of 140 labelled training texts and 110 development examples used for model validation during the develop-

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ment phase. For the final evaluation, the organizers released a larger unlabeled test set containing 27,972 Arabic text instances, together with genre definitions covering 74 specific genres under six broad genre categories.

3 System Overview

3.1 Data Processing

The dataset was inspected to ensure consistency among the text instances, genre labels, and genre definitions. Arabic text normalization was applied, including whitespace cleanup and normalization of common orthographic variants (Hendi, 2026). Aggressive preprocessing was avoided to preserve stylistic and linguistic cues that may be useful for genre classification.

3.2 Synthetic Data Preparation

To address the limited size of the official training set, we used GPT-4.1 mini to generate synthetic examples based on the official genre definitions, ensuring consistent broad and specific genre labels.

The generated data was cleaned and validated by removing malformed records and verifying label names and parent-child consistency. The final synthetic dataset contained 22,199 examples, with approximately 300 examples for most specific genres, and was combined with the 140 official training examples. As shown in Figure 1, this substantially increased the number of examples across broad genres, particularly the Informative category. The generation prompt and preprocessing scripts are available on GitHub.

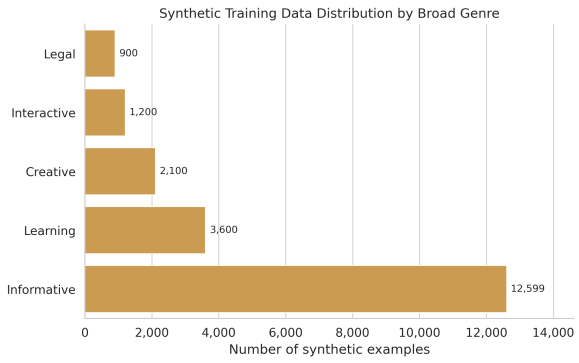


Figure 1: Distribution of the official and synthetic training examples across the broad genre categories.

4 Proposed Model

The proposed system follows a definition-guided hierarchical classification approach. Instead of treating genre labels only as categorical identifiers, the model represents both Arabic input texts and genre definitions in the same semantic embedding space. Each input text is first encoded using the pretrained encoder, producing a 768-dimensional text embedding. Similarly, each genre definition is encoded into a 768-dimensional definition embedding. A lightweight neural projection model is then trained to map text embeddings closer to their corresponding genre-definition embeddings while also learning the broad genre category through an auxiliary classification head.

The model consists of four main modules: the text and definition embedding module, the Conformer-based projection module, the broad-genre classification module, and the ensemble prediction module. Together, these modules allow the system to use semantic similarity between texts and label definitions while preserving the hierarchical relationship between broad and specific genres. The core processing pipeline, from multilingual text encoding to broad- and specific-genre prediction, is illustrated in Figure 2.

4.1 Text and Definition Embedding Module

The first module encodes both input texts and genre definitions using a pretrained multilingual MPNet-based sentence encoder (Kudo and Richardson, 2018; Reimers and Gurevych, 2019). The encoder maps each input text into a 768-dimensional semantic representation. Genre definitions are encoded using the same model, with each definition incorporating information from both the broad genre description and its corresponding specific genre definition. Sharing the encoder between texts and definitions ensures that both are represented within a common semantic embedding space, facilitating their subsequent comparison.

4.2 Conformer-Based Projection Module

The second module is a Conformer-based projection head that receives the 768-dimensional text embedding as input. As shown in Figure 2, the text embedding is transformed by a linear projection layer into a sequence of four 512-dimensional representations. A learnable positional embedding is then added to the projected sequence before it is passed through a stack of Conformer blocks.

Each Conformer block consists of two feed-forward modules (Gulati et al., 2020), a multi-head self-attention module, and a convolution module, connected through residual connections, as

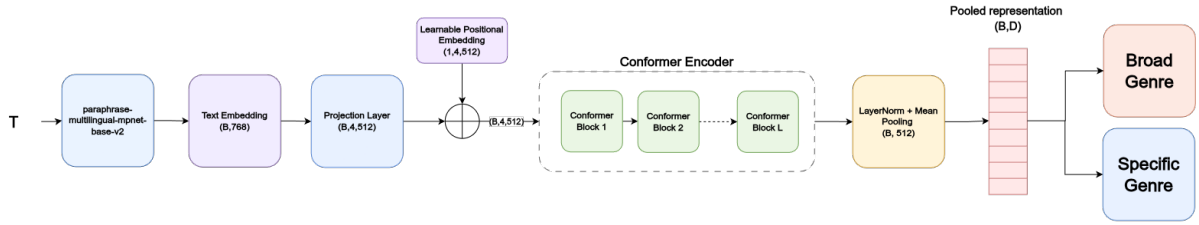


Figure 2: Core architecture of the proposed hierarchical genre classification model. The pretrained multilingual MPNet encoder produces a 768-dimensional text embedding, which is projected into four 512-dimensional representations, combined with learnable positional embeddings, processed by a stack of Conformer blocks, and pooled to generate broad- and specific-genre predictions.

illustrated in Figure 3. The self-attention module captures global relationships among the projected representations, whereas the convolution module captures local feature interactions. The two feed-forward modules further transform the representations before a final LayerNorm operation is applied. After the complete stack of Conformer blocks, LayerNorm and mean pooling produce a single 512-dimensional representation for downstream broad- and specific-genre prediction.

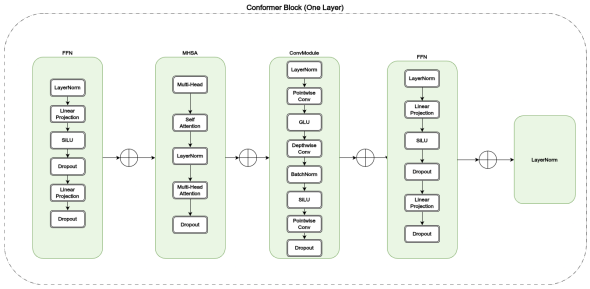


Figure 3: Internal architecture of a Conformer block used in the proposed projection module. Each block contains two feed-forward modules, a multi-head self-attention module, and a convolution module connected through residual connections, followed by a final LayerNorm operation.

4.3 Broad-Genre Classification Module

In addition to the projection output, the model includes an auxiliary broad-genre classification head. This module predicts the broad genre category of each text and provides hierarchical supervision during training. The broad-genre loss helps the model learn high-level genre distinctions, while the projection objective focuses on fine-grained specific genre alignment. During training, a balanced broad-genre loss was used to reduce the effect of category imbalance.

4.4 Ensemble Prediction Module

To improve robustness, the final system uses an ensemble of three independently trained models with random seeds **42**, **52**, and **62**. During inference, each model projects the input text representation and computes its cosine similarity with all specific-genre definition embeddings.

For each specific genre, the similarity scores from the three models are averaged, and the genre with the highest average score is selected as the final prediction. Its corresponding parent category is then used as the broad-genre prediction, ensuring hierarchical consistency between the two levels.

5 Experimental Setup

5.1 Data Splits

The official training set was used for supervised training together with the cleaned synthetic data. The development set was used during the development phase for validation and model selection. The final test set was used only for generating the official shared-task submission. No gold labels from the final test set were used during training.

5.2 Training Procedure

The model was trained on the combined official labelled and cleaned synthetic datasets using precomputed text embeddings. The projected text representation was aligned with the corresponding genre definition embedding using mean squared error, while an auxiliary broad-genre classifier was trained with weighted cross-entropy to address class imbalance. The overall objective was:

$$\mathcal{L} = \mathcal{L}_{proj} + \lambda \mathcal{L}_{broad},$$

where $\mathcal{L} * proj$ is the definition alignment loss, $\mathcal{L} * broad$ is the broad-genre classification loss, and $\lambda = 0.3$. Before computing $\mathcal{L}_{proj}$, the pooled

512-dimensional Conformer representation was projected back to the 768-dimensional definition-embedding space using a linear layer.

The model used four projected tokens, a hidden size of 512, one Conformer layer with four attention heads and a convolution kernel size of 15. Training used AdamW with a learning rate of 1×10^{-3}, weight decay of 1×10^{-4}, and a batch size of 32 for up to 180 epochs with early stopping. Three models were trained with random seeds 42, 52, and 62, and their outputs were combined during inference.

5.3 External Tools and Libraries

The system was implemented in Python using PyTorch, Hugging Face Transformers, and Sentence-Transformers. Additional libraries such as NumPy and scikit-learn were used for numerical processing, evaluation, and analysis. Synthetic data generation and preprocessing scripts are included in the project repository.

5.4 Evaluation Metrics

The official evaluation includes both broad-level and specific-level metrics. The main ranking metric is Hierarchical Macro-F1, which combines performance across the two hierarchical levels. The additional reported metrics are Specific Macro-F1, Specific Weighted-F1, Specific Accuracy, Broad Macro-F1, Broad Weighted-F1, and Broad Accuracy.

6 Results

6.1 Official Final Evaluation

Table 1 reports the official final evaluation results for our submitted system. Overall, the model performed substantially better at the broad-genre level than at the specific-genre level. It achieved a Broad Macro-F1 of 0.6288 and a Broad Accuracy of 0.7128, indicating that the model was relatively effective at identifying high-level communicative functions. However, performance decreased considerably for fine-grained classification, with a Specific Macro-F1 of 0.2511. This suggests that distinguishing between closely related specific genres remained challenging, particularly when subcategories shared similar vocabulary, writing style, or communicative purposes. Consequently, the final Hierarchical Macro-F1 of 0.4399 reflects this difference in performance across the two levels, combining relatively strong broad-genre classification with weaker discrimination among specific genres.

Metric	Score
Hierarchical Macro-F1	0.4399
Specific Macro-F1	0.2511
Specific Weighted-F1	0.2648
Specific Accuracy	0.2709
Broad Macro-F1	0.6288
Broad Weighted-F1	0.7163
Broad Accuracy	0.7128

Table 1: Official final evaluation results of the submitted system.

6.2 Discussion

The experiments showed that synthetic data generation improved coverage of the fine-grained label space, while longer training mainly benefited specific-genre prediction. Ensembling models trained with different random seeds also improved prediction stability.

The main errors occurred between semantically similar specific genres within the same broad category. This explains why broad-genre performance was stronger than specific-genre performance, as sibling genres often share similar vocabulary, style, and communicative purpose. The limited size of the official labelled training set and the reliance on synthetic data also contributed to these errors. Future work should therefore focus on improving discrimination between closely related specific genres.

7 Conclusion

This paper presented a definition-guided hierarchical system for Arabic genre classification using multilingual sentence embeddings, genre-definition representations, a Conformer-based projection module, an auxiliary broad-genre classifier, synthetic data augmentation, and seed-based ensembling. The final system achieved a Hierarchical Macro-F1 score of 0.4399 on the official evaluation set.

The main limitation is distinguishing fine-grained genres, particularly semantically similar sibling categories. Although synthetic data improved label coverage, it may not fully represent the diversity of real examples. Future work should explore stronger reranking, contrastive hard-negative training, improved synthetic data filtering, and hierarchical decoding.

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